PLUS or Minus?
The Effect of Graduate School Loans on Access, Attainment, and Prices

ONLINE APPENDIX: NOT FOR PUBLICATION

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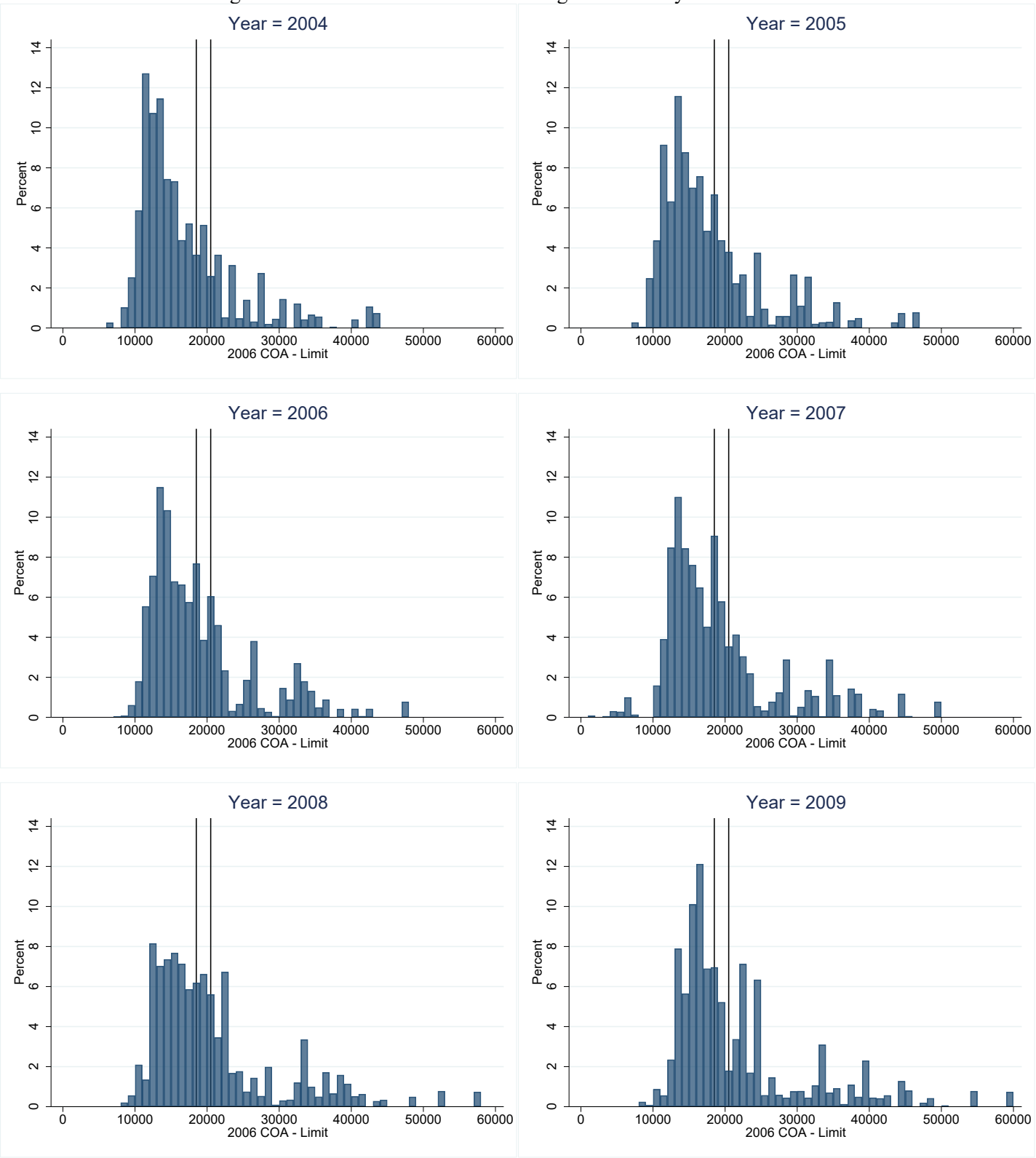
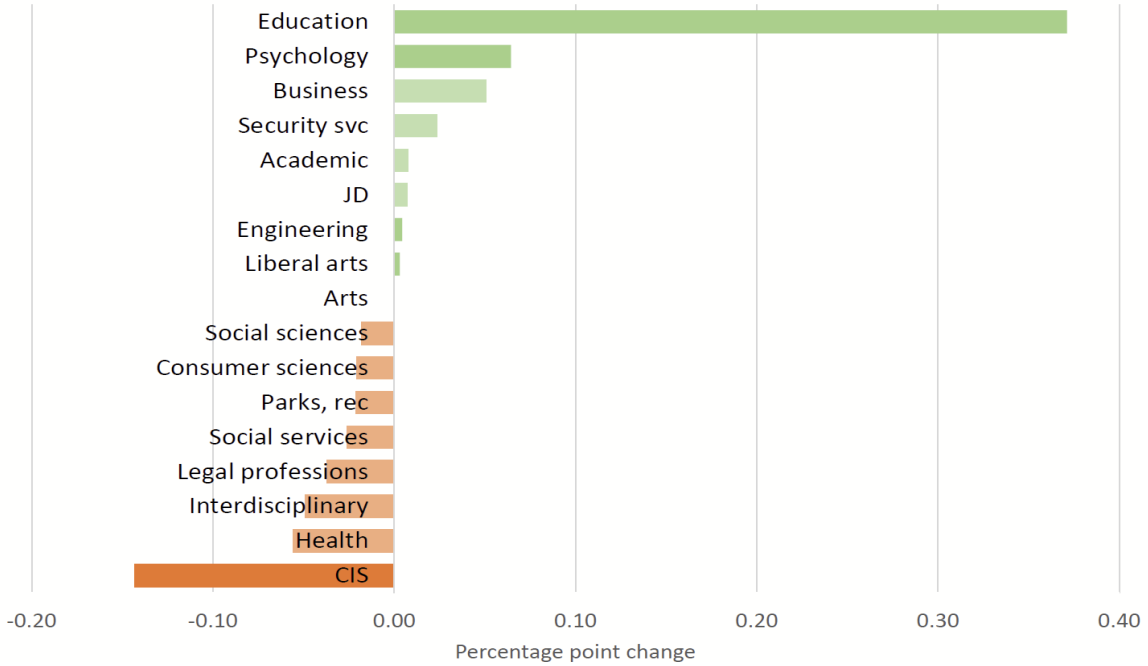
Appendix A: Additional Tables and Figures

Figure A.1: Distribution of Nominal Program Prices by Academic Year



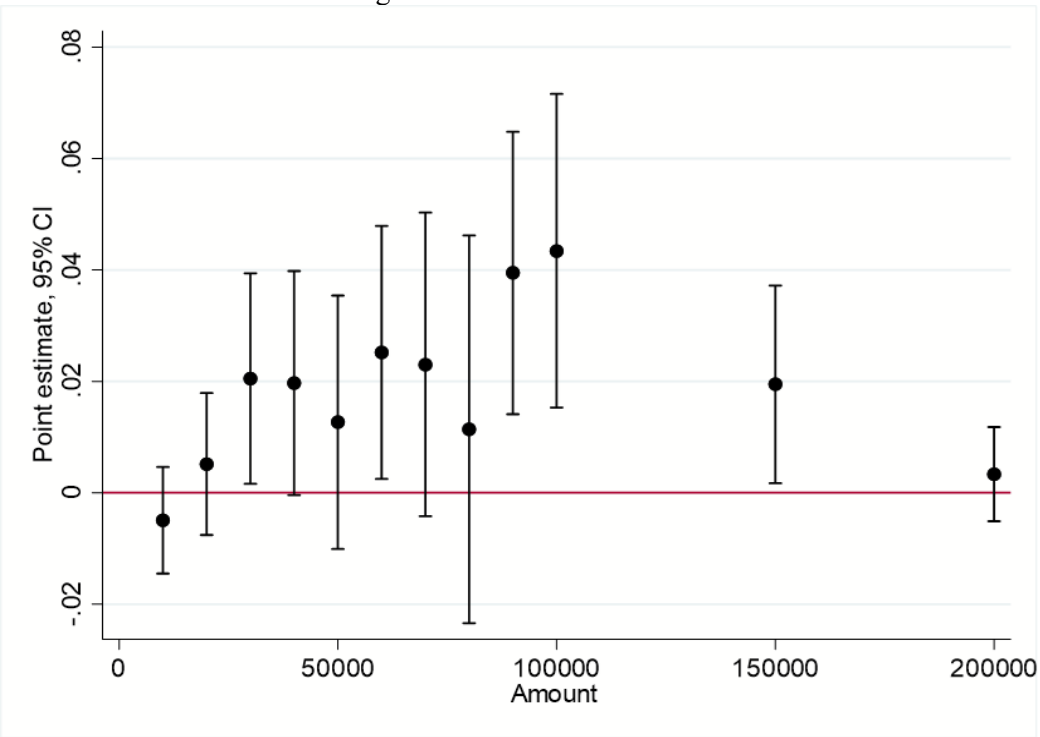
Notes: Each panel shows the distribution of the cost of attendance in a given academic year for graduate programs included in the enrollment analysis sample (graduate programs that had entering students in each of the 2004 through 2010 academic years and had at least 20 entering students in 2004 through 2006). The vertical lines represent the pre- and post-Grad PLUS Stafford Loan limits.

Figure A.2: Percentage-point change in new program fields post-Grad PLUS



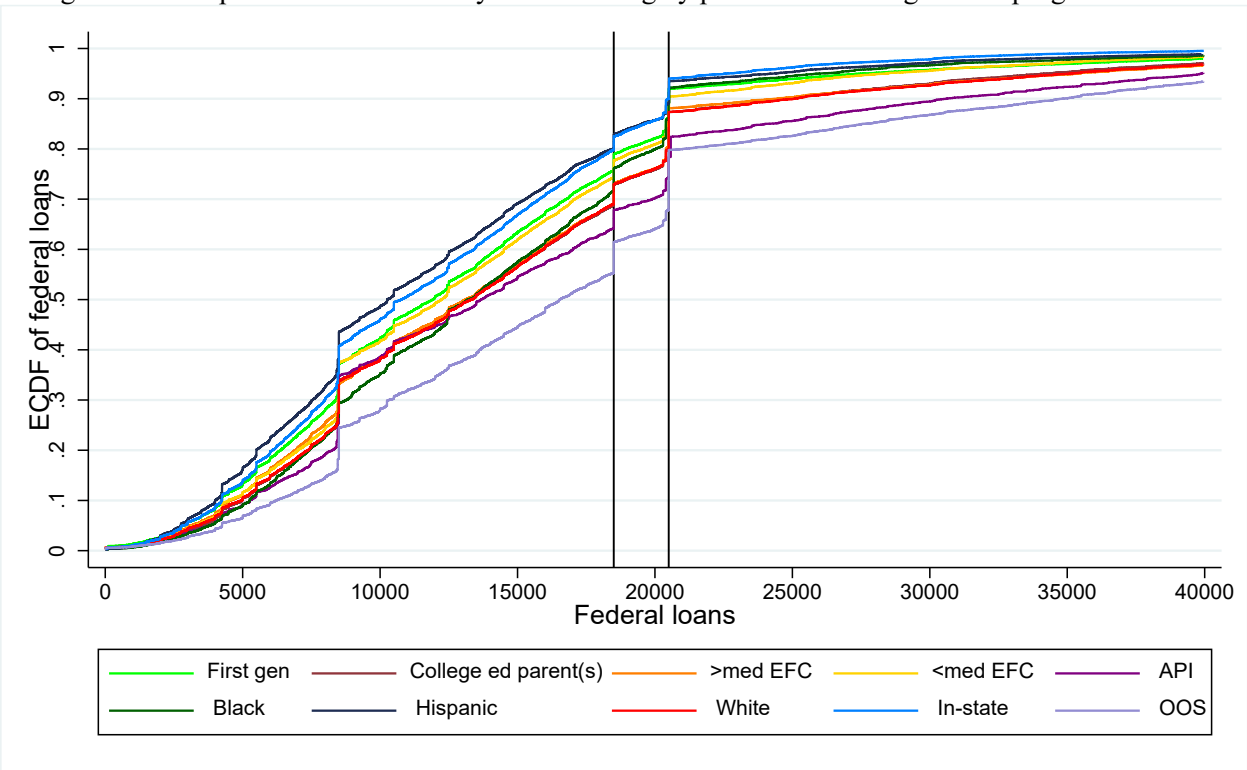
Notes: Each bar represents the difference in the percentage of new programs offered in a given field post-Grad PLUS (2007-2010) compared to pre-Grad PLUS (2004-2006).

Figure A.3: Estimated effects of higher loan limits due to Grad PLUS on constrained students' probability of borrowing different levels of cumulative debt



Notes: This figure plots point estimates and 95 percent confidence intervals from regressions of the outcome on the x-axis on an interaction between entry cohort and an indicator for being constrained (borrowing at the federal Stafford Loan limit). Regressions also include entry cohort and entry program fixed effects and age, indicators for race/ethnicity, gender, college educated parents and enrollment in fall and spring, and constrained. Standard errors are clustered at the program level. The outcome is an indicator for cumulative borrowing at or above the amount indicated on the x-axis, measured 6 years after entry.

Figure A.4: Empirical CDFs for first-year borrowing by post-Grad PLUS graduate program entrants



Notes: The sample includes students who borrowed in their first year of graduate school and belonged to the 2007 and 2008 entry cohorts (excluding students who entered health professions programs due to higher annual limits for these programs). The black vertical lines indicate the pre-Grad PLUS Stafford Loan limit ($18,500) and the post-Grad PLUS Stafford Loan limit ($20,500) for non-health professions programs.

Table A.1: Characteristics of entering graduate students in Texas data and NPSAS

	Pre Grad PLUS		Post Grad PLUS	
	(1) 2004 NPSAS	(2) Texas	(3) 2008 NPSAS	(4) Texas
Female	0.58	0.58	0.61	0.58
Race/ethnicity/nativity*				
Asian	0.10	0.06	0.10	0.06
Black	0.11	0.10	0.16	0.10
Hispanic	0.16	0.16	0.16	0.16
White	0.66	0.55	0.64	0.53
International student	0.07	0.13	0.08	0.13
Age (in yrs.)	31	30	31	29
Broad field of study**				
Education	0.29	0.21	0.28	0.20
Business	0.20	0.12	0.22	0.13
Health	0.10	0.12	0.11	0.13
Engineering	0.04	0.06	0.05	0.06
Law	0.05	0.05	0.04	0.04
Computer science	0.02	0.03	0.03	0.02
Arts	0.02	0.01	0.02	0.01
Academic doctoral	**	0.14	**	0.13
Borrowers	0.44	0.32	0.47	0.40
Constrained borrowers	0.15	0.10	0.14	0.16
Amount borrowed (2018$)				
Total loans	$9,915	$5,637	$10,543	$7,459
Federal Stafford loans	$8,462	$5,237	$8,180	$6,565
Federal PLUS loans	--	--	$951	$597
Federal Perkins loans	$216	$82	$122	$49
State loans	‡	$17	$21	$50
Private and other loans	$1,096	$289	$1,225	$198
Total grants (2018$)	$3,396	$734	$4,119	$1,551
Cost of attendance (2018$)	$27,331	$26,064	$29,448	$29,940

Notes: * Race/ethnicity for all students. In Texas data we do not observe race/ethnicity for international students.
** N/A, academic doctoral students are included in other categories. ‡ Didn't meet NPSAS reporting standards. Constrained borrowers are those with annual federal student loans at the statutory limit. These estimates are based on first time entering graduate students in the Texas data (the nationally representative sample of entering graduate students in the 2004 and 2008 waves of the National Postsecondary Student Aid Study (via PowerStats). The codes to retrieve 2004 results are: tykqbj, grklqp, bsagot, nqdzsk. The codes to retrieve 2008 results are: sznmjt, bagrvd, ngxiia.

Table A.2: Characteristics of programs by cost of attendance relative to pre-Grad PLUS loan limit

Predicted limit increase =	*(1) Greater than zero*	*(2) Zero*
Sector		
Public	0.757	0.889
Private nonprofit	0.147	0.013
Health-related	0.095	0.098
Field of study#		
Education	0.079	0.334
Academic	0.256	0.143
Business	0.128	0.139
Law	0.214	0.000
Allied health	0.079	0.098
Engineering	0.101	0.051
Social services	0.027	0.037
Computer science	0.026	0.017
Psychology	0.000	0.031
Library sciences	0.003	0.028
Architecture	0.022	0.006
Communications	0.013	0.004
Arts	0.002	0.004
STEM	0.133	0.100
STEM + medical degrees	0.176	0.191
Capacity		
Spread	0.276	0.319
Above median spread	0.399	0.541
Selectivity		
Acceptance rate	0.459	0.753
Below median acceptance rate	0.843	0.254
Size of entering cohort	404	309
Percent (0-100) entering students who are:		
API	6.2	5.7
Black	5.8	9.1
Hispanic	9.2	19.6
White	57.3	54.4
International	18.0	9.1
Men	52.0	36.8
First-generation	29.6	44.3
Constrained	22.7	6.8
Average EFC (2018$)	$9,141	$6,243
Average loans (2018$)		
PLUS	$14	$9
Stafford	$8,417	$5,944
Federal	$8,535	$6,049
State or private	$1,257	$195
Total	$9,688	$6,148

Notes: The sample includes programs in the enrollment analysis sample (see Figure A.1 notes for details) in the 2004 through 2006 academic years. The predicted limit increase is the difference in a program's 2006 cost of attendance and the 2006 Stafford Loan limit. STEM classification is based on the DHS definition. Spread is the difference between the largest and smallest entry cohort, divided by the average entry cohort size, in 2004 through 2006. The acceptance rate is the percentage of applicants accepted to the program; see Appendix B for additional details. API = Asian or Pacific Islander. EFC = expected family contribution.

Table A.3: Characteristics of constrained and unconstrained first-year borrowers

	Pre-Grad PLUS (2002-2006)		Post Grad PLUS (2007-2010)	
	(1) Unconstrained	(2) Constrained	(3) Unconstrained	(4) Constrained
Observations	54,802	23,630	52,751	33,145
A. Student characteristics				
Female	0.633	0.538	0.635	0.573
Race/ethnicity[1]				
API	0.067	0.055	0.074	0.059
Black	0.143	0.147	0.146	0.169
Hispanic	0.203	0.151	0.232	0.178
White	0.584	0.644	0.544	0.590
Age (in yrs)	28	28	28	29
Texas Resident	0.887	0.814	0.899	0.849
EFC (2018$)	$6,467	$10,910	$7,480	$10,790
B. Resources and cost of attendance				
Any loans				
PLUS	--	--	0.019	0.219
Private	0.036	0.161	0.039	0.041
Amount borrowed (2018$)				
Total loans	$15,689	$28,703	$14,655	$29,805
Federal Stafford Loans	$14,989	$25,708	$13,684	$24,979
Federal PLUS loans	--	--	$207	$3,923
Federal Perkins loans	$310	$230	$141	$127
State loans	$26	$132	$124	$159
Private and other loans	$313	$2,606	$498	$616
Total grants (2018$)	$2,278	$2,359	$2,747	$3,134
Earnings (2018$)	$21,663	$19,549	$26,175	$24,196
Cost of attendance (2018$)	$25,342	$38,236	$27,579	$40,061
Available resources (2018$	$14,288	$12,365	$15,995	$17,063
C. Program characteristics				
Broad field of study				
Education	0.214	0.142	0.220	0.171
Health	0.246	0.149	0.231	0.194
Business	0.105	0.100	0.139	0.142
Engineering	0.021	0.012	0.022	0.014
Law	0.034	0.322	0.033	0.211
Social Services	0.052	0.030	0.050	0.053
Computer science	0.010	0.008	0.010	0.005
Psychology	0.047	0.007	0.049	0.004
Library science	0.017	0.006	0.014	0.009
Academic doctoral	0.166	0.149	0.146	0.114
STEM field	0.079	0.035	0.079	0.044
STEM/med deg field	0.231	0.084	0.213	0.127
D. School characteristics				
Sector of institution				
Health-related	0.141	0.064	0.117	0.082
Flagship	0.135	0.167	0.091	0.107
R1	0.263	0.265	0.268	0.248
Other	0.461	0.504	0.525	0.563
Type of institution				
Public university	0.756	0.587	0.714	0.551
Private nonprofit	0.105	0.350	0.170	0.368
Health-related	0.139	0.063	0.116	0.081

Notes: 1. Among students with a non-missing race/ethnicity. The sample includes first-time graduate students in the entry cohorts indicated in column headings who borrowed in their first year of graduate school. API = Asian or Pacific Islander. EFC = expected family contribution. Constrained borrowers are students with federal loans at or above the Stafford Loan maximum, unconstrained borrowers are students who received student loans below this amount.

Table A.4: Characteristics of graduate programs by cost of attendance relative to Stafford Loan limit

Predicted limit increase =	*Greater than zero*		*Zero*	
	(1) Pre	(2) Post	(3) Pre	(4) Post
Size of entering cohort	404	375	309	282
Percent (0-100) entering students who are:				
API	6.2	7.3	5.7	6.0
Black	5.8	6.4	9.1	10.0
Hispanic	9.2	10.2	19.6	20.9
White	57.3	56.0	54.4	50.6
International	18.0	16.8	9.1	9.3
Men	52.0	51.8	36.8	36.7
First-generation	29.6	36.3	44.3	45.9
Constrained	22.7	24.5	6.8	11.7
Average EFC (2018$)	$9,141	$9,471	$6,243	$7,828
Average loans (2018$)				
PLUS	$14	$1,823	$9	$93
Stafford	$8,417	$8,461	$5,944	$6,979
Federal	$8,535	$10,364	$6,049	$7,125
State or private	$1,257	$419	$195	$249
Total	$9,688	$10,703	$6,148	$7,320
Federal loan limit (2018$)	$26,352	$36,353	$22,036	$23,737
Percent (0-100) with:				
PLUS loans	0	11	0	1
Federal loans	39	41	34	40
Loans from any source	39	42	34	40

Notes: The sample includes programs in the enrollment analysis sample (see Figure A.1 notes for details) in the 2004 through 2006 academic years. The predicted limit increase is the difference in a program's 2006 cost of attendance and the 2006 Stafford Loan limit. Pre is 2004-2006, post is 2007-2010. API = Asian or Pacific Islander.

Table A.5: Effect of projected and realized increases in federal loan limits on borrowing by entering graduate students

	Percent (0-100%) of entering students with:			Average amount borrowed:		
	(1) PLUS loans	(2) Federal loans	(3) Any loans	(4) PLUS loans	(5) Federal loans	(6) Total loans
A.OLS estimates						
Projected limit increase ($1k) x post	1.20	-0.21	-0.21	237	180	48
	(0.19)**	(0.10)*	(0.10)*	(31)**	(52)**	(42)
B. IV estimates						
Federal loan limit ($1k)	1.15	-0.21	-0.21	228	173	46
	(0.15)**	(0.09)*	(0.09)*	(23)**	(45)**	(37)
C. Pre-Grad PLUS mean	0.1	35.3	35.4	11	6812	7235

Notes: The sample includes programs in the enrollment analysis sample (see Figure A.1 notes for details; N = 2,380). Panel A displays point estimates from regressions of the percentage of entering students (0-100) who borrowed the indicated type of loan or the average (unconditional) amount borrowed by entering graduate students on an interaction between post-Grad PLUS entry cohort and the projected federal loan limit increases (see text for details). Panel B displays point estimates from instrumental variables models in which the interaction between the projected limit increase and the indicator for post-Grad PLUS serves as the excluded instrument for the realized federal loan limit (F-stat = 675). Panel C displays the mean of the dependent variable in 2004-2006. All specifications also include entry cohort and program fixed effects and observations are weighted by the size of the entering cohort. Robust standard errors, clustered at the program level, in parentheses; ** $p < 0.01$, * $p < 0.05$.

Table A.6: Robustness of estimated effects on program enrollment and composition

	(1)	Percent (0-100%) of entering students who are:						
	Enrollment	(2) Black	(3) Hispanic	(4) API	(5) White	(6) Internl	(7) Men	(8) First gen
A. Average COA 2004-2006								
Projected limit increase ($1k) x Po	-1.1	-0.02	0.01	0.03	0.08	-0.03	0.00	0.10
	(0.9)	(0.02)	(0.02)	(0.0149)+	(0.07)	(0.05)	(0.03)	(0.08)
Observations	2380	2380	2380	2380	2380	2380	2380	2376
B. Limit includes grant aid								
Projected limit increase ($1k) x Po	-1.1	-0.02	0.02	0.02	0.09	-0.04	-0.01	0.11
	(1.0)	(0.02)	(0.03)	(0.02)	(0.10)	(0.08)	(0.03)	(0.11)
Observations	2380	2380	2380	2380	2380	2380	2380	2376
C. Predicted FTFY COA								
Projected limit increase ($1k) x Po	-1.4	-0.02	-0.01	0.01	-0.11	0.11	-0.12	0.03
	(1.10)	(0.05)	(0.05)	(0.04)	(0.10)	(0.07)	(0.07)	(0.14)
Observations	2373	2373	2373	2373	2373	2373	2373	2369
D. Expand post-period to 2013								
Projected limit increase ($1k) x Po	-0.9	-0.06	0.33	-0.02	-0.09	-0.04	0.01	0.17
	(0.8)	(0.0230)**	(0.0772)**	(0.03)	(0.09)	(0.06)	(0.04)	(0.11)
Observations	3685	3685	3685	3685	3685	3685	3685	3685
E. 10 student minimum								
Projected limit increase ($1k) x Po	-0.9	-0.01	0.02	0.02	0.08	-0.03	-0.01	0.12
	(0.7)	(0.02)	(0.03)	(0.01)	(0.09)	(0.06)	(0.03)	(0.09)
Observations	2961	2961	2961	2961	2961	2961	2961	2961
F. Include HBCU programs								
Projected limit increase ($1k) x Po	-1.0	-0.01	0.02	0.02	0.07	-0.03	-0.002	0.13
	(0.8)	(0.02)	(0.03)	(0.02)	(0.09)	(0.06)	(0.03)	(0.09)
Observations	2443	2443	2443	2443	2443	2443	2443	2439
G. Exclude academic doctoral programs								
Projected limit increase ($1k) x Po	-0.9	-0.02	0.01	0.01	0.02	0.03	0.02	0.09
	(0.9)	(0.02)	(0.03)	(0.01)	(0.06)	(0.02)	(0.03)	(0.09)
Observations	2121	2121	2121	2121	2121	2121	2121	2117
H. Exclude law and psychology programs								
Projected limit increase ($1k) x Po	-0.1	-0.06	-0.05	0.12	0.36	-0.23	-0.11	0.34
	(1.3)	(0.05)	(0.06)	(0.04)**	(0.23)	(0.24)	(0.07)	(0.27)
Observations	2191	2191	2191	2191	2191	2191	2191	2187
I. Exclude bunchers								
Projected limit increase ($1k) x Po	-1.2	-0.01	0.02	0.02	0.07	-0.03	0.01	0.09
	(0.9)	(0.02)	(0.03)	(0.02)	(0.09)	(0.06)	(0.03)	(0.09)
Observations	2156	2156	2156	2156	2156	2156	2156	2152

Table A.6, cont.

	(1)	Percent (0-100%) of entering students who are:						
	Enrollment	(2) Black	(3) Hispanic	(4) API	(5) White	(6) Internl	(7) Men	(8) First gen
J. Unweighted								
Projected limit increase ($1k) x Po	-1.0	-0.04	-0.02	0.07	0.00	0.00	0.01	0.09
	(0.8)	(0.03)	(0.07)	(0.0328)*	(0.07)	(0.04)	(0.04)	(0.10)
Observations	2380	2380	2380	2380	2380	2380	2380	2380
K. Binary treatment								
Treat_p x Post	-10.3	-0.30	-0.34	0.82	2.51	-1.27	-0.15	5.11
	(12.5)	(0.32)	(0.38)	(0.34)+	(1.25)+	(1.21)	(0.63)	(1.73)*
Observations	2380	2380	2380	2380	2380	2380	2380	2376
L. Piecewise linear function in projected limit increase								
Projected limit increase ($1k) x Post								
x **1**[Projected limit increase <$5.	-1.2	-0.17	-0.17	0.24	0.68	-0.11	-0.06	2.62
	(3.6)	(0.093)+	(0.12)	(0.19)	(0.290)*	(0.30)	(0.25)	(0.689)**
x **1**[Projected limit increase ≥$5.	-1.0	-0.01	0.02	0.02	0.07	-0.03	0.00	0.10
	(0.8)	(0.02)	(0.03)	(0.02)	(0.09)	(0.06)	(0.03)	(0.09)
Test of equality (*p* -value)	0.954	0.063	0.095	0.231	0.047	0.805	0.810	0.000
Observations	2380	2380	2380	2380	2380	2380	2380	2376
M. Separate treatment effects for bunchers								
Projected limit increase ($1k) x Po	-0.3	0.03	0.04	0.05	-0.09	-0.02	0.02	0.06
	(0.8)	(0.05)	(0.05)	(0.05)	(0.13)	(0.08)	(0.06)	(0.16)
1[Buncher] x Post	-0.5	-0.03	-0.02	-0.03	0.13	-0.01	-0.01	0.05
	(0.4)	(0.04)	(0.03)	(0.04)	(0.08)	(0.05)	(0.04)	(0.10)
Observations	2380	2380	2380	2380	2380	2380	2380	2376

Notes: Unless otherwise noted, the sample includes graduate programs students in each of the 2004 through 2010 entry cohorts and had at least 20 entering students who filed a FAFSA in 2004 through 2006. Panel D includes observations of graduate programs through 2013. Panel E sample includes graduate programs students in each of the 2004 through 2010 entry cohorts and had at least 10 entering students who filed a FAFSA in 2004 through 2006. Panel F sample includes programs in HBCUs. Panels G and H samples exclude the indicated programs. Panel I sample excludes programs with a 2006 COA just below the 2006 Stafford Loan limit ("bunchers"). Each panel contains point estimates from separate regressions of first-year enrollment or percentage of entering students (0-100) with the given demographic characteristic on an interaction between post-Grad PLUS entry cohort and treatment. All specifications also include entry cohort and program fixed effects. In Panel A, the treatment is the projected limit increase based on a program's average cost of attendance (COA) between 2004 and 2006. In Panel B, the treatment is the projected limit increase based on the 2006 COA less student grant aid. In Panel C, the treatment is the projected limit increase based on the predicted COA for a full-time, full-year student in 2006. In Panels D through J, the treatment is the projected federal loan limit increase based on the program's 2006 COA. In Panel K, the treatment is an indicator for whether the program had a 2006 COA above the Stafford Loan limit. In Panel L, the treatment is a piecewise linear spline in projected limit increase with an indicator for the median increase among programs with greater than 0 predicted limit increase. In Panel M, the treatment is the projected federal loan limit increase based on the program's 2006 COA and an indicator for having a 2006 COA just below the 2006 Stafford Loan limit ("bunchers"). All specifications also include entry cohort and program fixed effects. Columns (2) through (8) estimates weighted by the size of the 2006 entering cohort except where noted. Robust standard errors, clustered at the program level, in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

Table A.7: Characteristics of existing and new programs before and after Grad PLUS

	Existing programs		New programs	
	(1) Pre	(2) Post	(3) Pre	(4) Post
A. Student characteristics				
Male	0.423	0.415	0.524	0.365
Race/ethnicity				
White	0.536	0.514	0.461	0.433
Black	0.090	0.102	0.073	0.153
Hispanic	0.154	0.166	0.149	0.280
API	0.059	0.068	0.069	0.046
URM	0.248	0.273	0.227	0.440
Age	31	31	31	33
College educated parent	0.593	0.560	0.644	0.432
First generation student	0.407	0.440	0.356	0.568
EFC	$7,303	$8,655	$6,371	$7,511
B. Student financial aid (averages)				
Any loans	0.327	0.385	0.281	0.351
Borrowing at Stafford limit	0.077	0.119	0.106	0.106
Total loans	$6,706	$7,852	$4,728	$5,632
Federal	$6,484	$7,622	$4,649	$5,614
State	$13	$19	$7	$2
Private	$208	$211	$72	$15
Total grants	$680	$1,228	$450	$912
C. Program characteristics				
COA	$25,383	$27,275	$24,389	$23,175
COA percentile rank				
In year	0.50	0.50	0.50	0.35
In school and year	0.61	0.61	0.34	0.66
D. School characteristics				
Health-related	0.116	0.127	0.113	0.030
Flagship	0.182	0.164	0.261	0.043
R1	0.322	0.309	0.208	0.143
Non-R1	0.380	0.400	0.418	0.784
HBCU	0.021	0.020	0.024	

Notes: The sample includes graduate programs in public or health-related institutions with enrollment between 2003 and 2010. New programs are those that did not have enrollment in prior academic years. Pre = Pre-Grad PLUS (2003-2006); Post = Post-Grad PLUS (2007-2010). Some cells are suppressed due to small underlying sample size. Private nonprofit institutions are excluded (only a subset of programs of study are observed). API = Asian or Pacific Islander. URM = underrepresented minority (Black, Hispanic, or Native American/American Indian). EFC = expected family contribution. COA = cost of attendance.

Table A.8: Effect of projected and realized increases in federal loan limits on enrollment and the composition of entering graduate students, school-level estimates

	(1) Enrollment	Percent (0-100%) of entering students who are:						
		(2) Black	(3) Hispanic	(4) API	(5) White	(6) Internl	(7) Men	(8) First gen
A.OLS estimates								
Projected limit increase ($1k) x post	-2.5	-0.04	-0.02	-0.01	0.03	0.11	-0.09	0.05
	(3.1)	(0.02)+	(0.02)	(0.02)	(0.08)	(0.08)	(0.042)*	(0.09)
B. IV estimates								
Federal loan limit ($1k)	-2.2	-0.04	-0.02	-0.01	0.03	0.10	-0.08	0.05
	(2.5)	(0.02)*	(0.01)	(0.02)	(0.06)	(0.07)	(0.03)*	(0.07)
95% CI	[-7.0, 2.6]	[-0.07, -0.001]	[-0.04, 0.01]	[-0.04, 0.02]	[-0.10, 0.15]	[-0.03, 0.22]	[-0.14, -0.01]	[-0.02, 0.11]
C. Pre-Grad PLUS mean	817.8	9.5	15.5	5.6	54.3	11.7	42.5	39.1

Notes: The sample includes public and nonprofit higher education institutions with graduate student enrollment in each of the 2004 through 2010 academic years (N = 446). Panel A displays point estimates from regressions of first-year enrollment or percentage of entering students (0-100) with the given demographic characteristic on an interaction between post-Grad PLUS entry cohort and the projected federal loan limit increase in the institution (see text for details). Panel B displays point estimates from instrumental variables models in which the interaction between the projected limit increase and the indicator for post-Grad PLUS serves as the excluded instrument for the realized federal loan limit (col. (1) F-stat = 65; col (2) – (8) F-stat = 122). Panel C displays dependent variable means, measured in 2004-2006. All specifications also include entry cohort and institution fixed effects. Columns (2) through (8) estimates weighted by the size of the number of entering graduate students at baseline. API = Asian or Pacific Islander. Robust standard errors, clustered at the program level, in parentheses; * $p < 0.05$, + $p < 0.1$.

Table A.9: Characteristics of programs by quartile of baseline percent of students who were constrained

	Q1 (<2.3%)	Q2 (2.3-5.6%)	Q3 (5.6-17.9%)	Q4 (≥ 17.9%)
Average enrollment	134	139	132	295
Percent of students who are:				
API	0.11	0.07	0.04	0.09
Black	0.08	0.09	0.13	0.13
Hispanic	0.21	0.13	0.15	0.11
White	0.50	0.60	0.56	0.62
Men	0.41	0.38	0.37	0.49
First generation	0.39	0.42	0.39	0.32
Average age	29.5	31.6	31.7	27.5
Average EFC	$6,412	$7,343	$9,185	$7,062
COA	$30,505	$26,030	$27,684	$45,996
FTFY COA	$21,614	$20,123	$22,298	$30,756
Net price	$28,627	$24,621	$26,441	$42,410
Percent with any loans	0.50	0.40	0.41	0.75
Average loans				
Grad PLUS	$60	$34	$160	$2,005
Stafford	$12,685	$8,272	$7,229	$21,189
State or private	$643	$193	$127	$1,589
Total	$13,638	$8,631	$7,574	$25,166
Average grants	$1,552	$1,121	$847	$3,277
Average tuition waiver	$325	$287	$395	$309
Sector of institution				
Flagship	0.122	0.084	0.283	0.148
R1	0.207	0.417	0.300	0.165
Other	0.376	0.399	0.307	0.502
Health-related	0.295	0.100	0.110	0.185
HBCU	0	0	0.055	0.077
STEM field of study	0.147	0.045	0.035	0.026
STEM or medical degree	0.513	0.147	0.075	0.145
Acceptance rate	0.661	0.669	0.665	0.340
Spread	0.274	0.309	0.336	0.177

Notes: The sample includes programs in the pricing analysis sample, which is a balanced panel graduate programs with enrollment in the 2003 through 2010 academic years and at least 20 federal aid recipients enrolled per year, on average, between 2003 and 2006. Baseline percent constrained is the percent of all students who had federal loans at the Stafford Loan maximum in 2003 through 2006. All dollar amounts adjusted for inflation using CPI-U and reported in 2018$. API = Asian or Pacific Islander, EFC = expected family contribution, COA = cost of attendance, FTFY COA = predicted COA for a full-time, full-year student. STEM classification based on the DHS definition. Spread is a measure of a program's capacity to expand, equal to the difference between the largest and smallest entering cohort in 2003 through 2006, divided by the average entering cohort size between 2003 and 2006.

Table A.10: Graduate program prices and characteristics before and after Grad PLUS

	(1) Pre	(2) Post
Percent of students who are:		
API	0.07	0.08
Black	0.10	0.11
Hispanic	0.15	0.16
White	0.58	0.56
Men	0.41	0.41
First gen	0.37	0.38
Average age	30.2	29.9
Average EFC	$7,136	$7,862
COA	$31,104	$34,084
FTFY COA	$21,308	$26,130
Net price	$29,566	$31,555
Percent with any loans	0.49	0.54
Average loans		
PLUS	$43	$1,096
Stafford	$11,933	$12,807
State or private	$710	$572
Total	$12,953	$14,620
Average grants	$1,347	$2,062
Average tuition waiver	$191	$467
Observations	1,168	1,168

Notes: The sample includes programs in the pricing analysis sample (see Figure A.9 notes for details). Pre = Pre-Grad PLUS (2003-2006), Post = Post-Grad PLUS (2007-2010). All dollar amounts adjusted for inflation using CPI-U and reported in 2018$. API = Asian or Pacific Islander, First gen = first generation student (no college educated parent), EFC = expected family contribution, COA = cost of attendance, FTFY COA = predicted COA for a full-time, full-year student.

Table A.11: Robustness of estimated effects on program-level loans and prices

	(3) COA	(4) FTFY COA	(1) Grad PLUS loans	(2) Total Fed. loans	(5) Grants	(6) Net price
A. Minimum program enrollment = 10						
% constrained*Post	66.9	63.4	88.0	60.7	25.6	39.3
	(23.5)**	(18.9)**	(18.4)**	(31.8)+	(15.1)+	(33.5)
Observations	3104	3104	3104	3104	3104	3104
B. Minimum program enrollment = 30						
% constrained*Post	66.1	62.4	88.2	60.7	25.6	38.6
	(23.5)**	(19.0)**	(18.5)**	(32.1)+	(15.3)	(33.6)
Observations	1992	1992	1992	1992	1992	1992
C. Exclude academic programs						
% constrained*Post	63.8	61.0	88.1	57.8	27.2	33.9
	(23.7)**	(19.0)**	(18.6)**	(32.3)+	(15.3)+	(33.8)
Observations	2040	2040	2040	2040	2040	2040
D. Exclude law and psychology programs						
% constrained*Post	75.2	92.7	24.7	35.0	-1.8	76.0
	(72.1)	(59.1)	(4.9)**	(37.6)	(3.6)	(74.8)
Observations	2096	2096	2096	2096	2096	2096
E. Consistent sample definition for inclusion in financial aid data						
% constrained*Post	66.1	62.6	90.3	62.4	26.3	37.0
	(23.4)**	(18.9)**	(19.1)**	(33.1)+	(15.3)+	(33.3)
Observations	2336	2336	2336	2336	2336	2336
F. Percent constrained based on 2006 only						
% constrained*Post	58.7	53.7	77.2	56.7	21.7	35.5
	(22.5)*	(19.7)**	(17.0)**	(26.6)*	(12.8)+	(29.5)
Observations	2336	2336	2336	2336	2336	2336
G. Constrained def does not account for partial year enrollment						
% constrained*Post	64.7	62.2	84.0	59.1	24.8	38.2
	(21.9)**	(17.7)**	(17.6)**	(30.0)+	(14.2)+	(31.1)
Observations	2336	2336	2336	2336	2336	2336
H. Constrained def based on all loans, accounts for partial year enrollment						
% constrained*Post	73.0	68.1	94.0	63.2	28.3	41.9
	(25.7)**	(20.2)**	(20.0)**	(35.4)+	(16.5)+	(36.9)
Observations	2336	2336	2336	2336	2336	2336
I. Constrained def based on all loans, not accounting for partial year enrollment						
% constrained*Post	71.7	67.0	91.1	62.8	27.3	41.9
	(24.3)**	(19.1)**	(19.0)**	(33.4)+	(15.6)+	(34.7)
Observations	2336	2336	2336	2336	2336	2336

Table A.11, cont.

	(3) COA	(4) FTFY COA	(1) Grad PLUS loans	(2) Total Fed. loans	(5) Grants	(6) Net price
J. Discrete treatment (above median % constrained)						
1[% cons > median]*Post	1048.0	1362.2	1955.7	1221.7	409.4	379.9
	(1013.7)	(808.2)+	(549.5)**	(824.8)	(294.4)	(1006.1)
Observations	2336	2336	2336	2336	2336	2336
K. Trinary (excluded category is % constrained < 5%)						
1[% cons in [5, 13)]*Post	-658	-123	207	-499	-50	-894
	(916)	(776)	(158)	(296)+	(100)	(870)
1[% cons ≥ 13]*Post	2479	2553	3489	2316	839	1471
	(1207)*	(1026)*	(857)**	(1180)+	(490)+	(1328)
Observations	2336	2336	2336	2336	2336	2336
L. Unweighted						
% constrained*Post	55.0	59.8	63.6	24.8	14.5	38.1
	(21.5)*	(17.5)**	(14.5)**	(23.3)	(12.7)	(27.4)
Observations	2336	2336	2336	2336	2336	2336

Notes: Unless otherwise noted, the sample includes a balanced panel graduate programs with enrollment in the 2003 through 2010 academic years and at least 20 federal aid recipients enrolled per year, on average, between 2003 and 2006. Panel A only requires 10 federal aid recipients enrolled per year, on average. Panel B requires 30 federal aid recipients enrolled per year, on average. Panels C and D exclude the indicated programs. Panel E imposes excludes students who did not receive aid requiring a needs calculation when constructing program-level prices and grant aid outcomes (see Appendix B for additional details). Each panel contains point estimates from separate regressions of the outcome indicated in the column heading on an interaction between post-Grad PLUS academic year and exposure to Grad PLUS. All specifications also include academic year and program fixed effects. In panels A through E and J through L, exposure to Grad PLUS equals the percent of students with federal Loans at or above the Stafford maximum (adjusted for part-year enrollment), on average, between 2004 and 2006. In Panel F, the same definition of constrained is used but exposure is only based on 2006 percent constrained. In Panel G, percent constrained is measured using total borrowing and does not account for part-year enrollment. In Panel H and I, percent constrained is the share of students with total loans at or above the Stafford, on average, between 2004 and 2006, where the definition of constrained is adjusted (H) or is not-adjusted (I) for part-year enrollment. In Panel J, exposure to Grad PLUS is measured by a binary variable indicating that the program had above median baseline percent constrained. In Panel K, exposure to Grad PLUS is measured by indicates for terciles of baseline percent constrained. Observations are weighted by baseline program size except in Panel L. Net price equals COA minus grants and tuition waivers. Robust standard errors, clustered at the program level, in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

Table A.12: Estimated effects on program-level loans and prices, simulated instrument treatment

	(1) Grad PLUS loans	(2) Total Fed. loans	(3) COA	(4) FTFY COA	(5) Grants	(6) Net price
% constrained×Limit Increase×Post	313.6	203.6	180.3	183.6	92.9	81.4
	(51.6)**	(110.1)+	(83.5)*	(54.9)**	(54.7)+	(112.8)
Observations	2336	2336	2336	2336	2336	2336

Notes: The sample includes programs in the pricing analysis sample (see Figure A.9 notes for details). Point estimates from regressions of average loans per student from the specified source on the triple interaction between the percentage of students who were constrained at baseline, the projected limit increases, and post-Grad PLUS (see text for definitions). Regressions also include academic year and program fixed effects. COA = cost of attendance. Net price equals COA minus grants and tuition waivers. Robust standard errors, clustered at the program level, in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

Table A.13: IV estimates of the effect of federal loans on program price using simulated instrument

	(1) COA	(2) FTFY COA	(3) Net price
Federal loans	0.885	0.902	0.400
	(0.486)+	(0.432)*	(0.400)
Observations	2336	2336	2336

Notes: The sample includes programs in the pricing analysis sample (see Figure A.9 notes for details). Point estimates from instrumental variables models in which average federal student loans per student is the endogenous regressor and the triple interaction between the percent of students who were constrained at baseline, the projected limit increase, and post-Grad PLUS (see text for definition) is the excluded instrument. Regressions also include academic year and program fixed effects. COA = cost of attendance. Net price equals COA minus grants and tuition waivers. Robust standard errors, clustered at the program level, in parentheses; * $p < 0.05$, + $p < 0.1$.

Table A.14: Characteristics of constrained and unconstrained borrowers by entry cohort

	Unconstrained borrowers		Constrained borrowers	
Entry cohort =	2002-2004	2005-2008	2002-2004	2005-2008
Demographics (measured at college entry)				
Female	0.629	0.638	0.529	0.552
Race/ethnicity				
API	0.068	0.065	0.052	0.055
Black	0.135	0.141	0.131	0.154
Hispanic	0.192	0.209	0.137	0.154
White	0.573	0.546	0.623	0.587
Texas resident	0.884	0.895	0.801	0.834
Age	27	28	28	28
Financial aid received in entry year (2018$)				
Grants	$2,110	$2,651	$2,154	$3,180
Federal Stafford loans	$15,179	$14,482	$25,785	$25,398
Federal PLUS loans	$35	$117	$13	$2,076
Federal Perkins loans	$340	$242	$248	$199
State	$28	$64	$108	$179
Private	$326	$394	$2,362	$1,781
Total loans	$15,908	$15,299	$28,516	$29,634
EFC (2018$)	$6,137	$7,182	$11,316	$10,736
COA (2018$)	$24,644	$26,968	$36,347	$40,374
Number of students	30,506	49,051	10,655	27,645

Notes: The sample includes first-time graduate students in the 2002 through 2008 entry cohorts who borrowed in their first year of enrollment. API = Asian or Pacific Islander. EFC = expected family contribution. COA = cost of attendance.

Table A.15: Placebo estimates

	(1) API	(2) Black	(3) Hispanic	(4) URM	(5) Age
Constrained x treated cohort	0.002	0.005	0.009	0.013	0.011
	(0.004)	(0.006)	(0.006)	(0.009)	(0.118)
Dep var mean (cons, pre)	0.052	0.131	0.137	0.274	28.030
	(6) Male	(7) College ed parent	(8) In-state	(9) EFC	(10) Index
Constrained x treated cohort	-0.002	-0.008	0.003	-1307	-0.001
	(0.008)	(0.009)	(0.018)	(551)*	(0.001)
Dep var mean (cons, pre)	0.471	0.648	0.815	$11,316	0.804

Notes: The sample includes first-time graduate students in the 2002 through 2008 entry cohorts who borrowed in their first year of enrollment. Point estimates from regressions of the baseline characteristic indicated in the column heading on an interaction between an indicator for being constrained (borrowing at the federal Stafford Loan limit) and an indicator for belonging to a treated entry cohort (academic year 2005 or later). Regressions also include entry cohort and entry program fixed effects. API = Asian or Pacific Islander. URM = underrepresented minority. EFC = expected family contribution (adjusted for inflation using CPI-U and reported in 2018$). Index is a linear prediction from a regression of the probability of earning a graduate degree within 10 years of entry on all baseline characteristics. Robust standard errors, clustered at the program level, in parentheses; * $p < 0.05$.

Table A.16: Effects of Grad PLUS on constrained students' receipt of specific degrees

Years since entry =	1	2	3	4	5	6	7	8	9	10
A. Master's degree										
Constrained x treated cohort	-0.005	0.009	0.003	0.0004	0.0004	0.001	0.002	0.001	0.001	0.0002
	(0.014)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)
Dep var mean (cons, pre)	0.271	0.438	0.488	0.509	0.519	0.526	0.530	0.534	0.538	0.540
B. Professional degree										
Constrained x treated cohort	‡	-0.006	0.001	0.004	0.006	0.006	0.006	0.006	0.007	0.007
	‡	(0.009)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Dep var mean (cons, pre)	‡	0.239	0.285	0.293	0.296	0.298	0.299	0.300	0.301	0.301
C. Doctoral degree										
Constrained x treated cohort	‡	‡	‡	‡	0.002	0.004	0.005	0.005	0.005	0.004
	‡	‡	‡	‡	(0.003)	(0.004)	(0.005)	(0.005)	(0.005)	(0.005)
Dep var mean (cons, pre)	‡	‡	‡	‡	0.013	0.017	0.020	0.023	0.026	0.029

Notes: The sample includes first-time graduate students in the 2002 through 2008 entry cohorts who borrowed in their first year of enrollment. Point estimates from regressions of the probability of the type of degree received (indicated in the panel heading) by the specified number of years since entry on an interaction between an indicator for being constrained (borrowing at the federal Stafford Loan limit) and an indicator for belonging to a treated cohort (academic year 2005 and later). Regressions also include entry cohort and entry program fixed effects and age, indicators for race/ethnicity, gender, college educated parents and enrollment in fall and spring, and constrained. Robust standard errors, clustered at the program level, in parentheses.

‡ results suppressed

Table A.17: Effects of Grad PLUS on constrained students' receipt of specific degrees by field of study

Years since entry =	2	3	4	5	6	7	8	9	10
A. Law degree (JD)									
Constrained x treated cohort	-0.010	0.001	0.003	0.004	0.005	0.005	0.005	0.005	0.005
	(0.007)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Dep var mean (cons, pre)	0.227	0.246	0.250	0.252	0.253	0.253	0.253	0.254	0.254
B. MBA									
Constrained x treated cohort	-0.010	-0.009	-0.008	-0.008	-0.008	-0.007	-0.007	-0.007	-0.007
	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)
Dep var mean (cons, pre)	0.145	0.151	0.154	0.155	0.156	0.157	0.158	0.158	0.159
C. Education master's degree									
Constrained x treated cohort	0.004	0.003	0.003	0.001	0.001	0.001	0.001	0.001	0.001
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Dep var mean (cons, pre)	0.073	0.085	0.091	0.095	0.098	0.100	0.102	0.103	0.104
D. Other health master's degree (excl. MPH and nursing)									
Constrained x treated cohort	0.004	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Dep var mean (cons, pre)	0.031	0.035	0.036	0.036	0.037	0.037	0.037	0.037	0.037
E. Accounting master's degree									
Constrained x treated cohort	0.0003	0.0002	-0.001	-0.0001	-0.0004	-0.0002	-0.0002	0.0001	0.00003
	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)
Dep var mean (cons, pre)	0.027	0.028	0.028	0.029	0.029	0.029	0.029	0.029	0.029
F. Chiropractic degree									
Constrained x treated cohort	-0.002	0.001	0.002	0.002	0.002	0.002	0.002	0.002	0.002
	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Dep var mean (cons, pre)	0.012	0.025	0.026	0.026	0.027	0.027	0.027	0.027	0.027
G. Nursing									
Constrained x treated cohort	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002
	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Dep var mean (cons, pre)	0.021	0.023	0.024	0.024	0.024	0.024	0.024	0.024	0.025
H. Medical degree (MD or DO)									
Constrained x treated cohort	‡	-0.0004	-0.001	-0.001	-0.001	-0.0004	-0.0004	-0.001	-0.001
	‡	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Dep var mean (cons, pre)	‡	0.010	0.013	0.013	0.014	0.014	0.014	0.015	0.015
I. Master's in Social Work									
Constrained x treated cohort	0.002	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003
	(0.001)+	(0.001)*	(0.001)*	(0.001)*	(0.001)*	(0.001)*	(0.001)*	(0.001)*	(0.001)*
Dep var mean (cons, pre)	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012
J. Other business master's degree (excl. MBA and accounting)									
Constrained x treated cohort	‡	-0.0003	-0.001	-0.002	-0.002	-0.002	-0.001	-0.002	-0.002
	‡	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Dep var mean (cons, pre)	‡	0.010	0.011	0.011	0.011	0.011	0.011	0.012	0.012
K. Engineering master's degree									
Constrained x treated cohort	‡	-0.001	-0.001	-0.0002	0.0002	0.0002	0.0002	0.0003	0.0004
	‡	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Dep var mean (cons, pre)	‡	0.010	0.011	0.011	0.011	0.011	0.011	0.011	0.011

Notes: The sample includes first-time graduate students in the 2002 through 2008 entry cohorts who borrowed in their first year of enrollment. Point estimates from regressions of the probability of the type of degree received (indicated in the panel heading) by the specified number of years since entry on an interaction between an indicator for being constrained (borrowing at the federal Stafford Loan limit) and an indicator for belonging to a treated cohort (academic year 2005 and later). Regressions also include entry cohort and entry program fixed effects and age, indicators for race/ethnicity, gender, college educated parents and enrollment in fall and spring, and constrained. Robust standard errors, clustered at the program level, in parentheses; * $p < 0.05$, + $p < 0.1$.

‡ results suppressed

Table A.18: Effects of Grad PLUS on constrained students' receipt of academic master's degrees

Years since entry =	2	3	4	5	6	7	8	9	10
A. Arts or humanities master's degree (academic)									
Constrained x treated cohort	0.002	0.003	0.003	0.004	0.004	0.004	0.005	0.004	0.004
	(0.005)	(0.005)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
Dep var mean (cons, pre)	0.020	0.025	0.027	0.028	0.029	0.029	0.029	0.030	0.030
B. Social science master's degree (academic)									
Constrained x treated cohort	0.002	0.002	0.0004	0.0002	0.0002	0.0004	0.001	0.001	0.001
	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Dep var mean (cons, pre)	0.010	0.013	0.015	0.016	0.016	0.017	0.017	0.017	0.017
C. Math or science master's degree (academic)									
Constrained x treated cohort	‡	0.002	0.002	0.003	0.003	0.003	0.002	0.002	0.002
	‡	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Dep var mean (cons, pre)	‡	0.011	0.011	0.012	0.012	0.012	0.012	0.013	0.013

Notes: The sample includes first-time graduate students in the 2002 through 2008 entry cohorts who borrowed in their first year of enrollment. Point estimates from regressions of the probability of the type of degree received (indicated in the panel heading) by the specified number of years since entry on an interaction between an indicator for being constrained (borrowing at the federal Stafford Loan limit) and an indicator for belonging to a treated cohort (academic year 2005 and later). Regressions also include entry cohort and entry program fixed effects and age, indicators for race/ethnicity, gender, college educated parents and enrollment in fall and spring, and constrained. Robust standard errors, clustered at the program level, in parentheses.
‡ results suppressed

Table A.19: Effects of Grad PLUS on constrained students' time to degree

	(1) Time to degree
Constrained x treated cohort	0.027
	(0.034)
Dep var mean (cons, pre)	2.06

Notes: The sample includes first-time graduate students in the 2002 through 2008 entry cohorts who borrowed in their first year of enrollment and received a graduate degree within 10 years of entry. Point estimates from regressions of the number of years between entry and degree receipt on an interaction between an indicator for being constrained (borrowing at the federal Stafford Loan limit) and an indicator for belonging to a treated cohort (academic year 2005 and later). Regressions also include entry cohort and entry program fixed effects and age, indicators for race/ethnicity, gender, college educated parents and enrollment in fall and spring, and constrained. Robust standard errors, clustered at the program level, in parentheses.

Table A.20: Effects of Grad PLUS on constrained students' available in-school resources and predicted loan payments

	Available resources in school		(3) Loan payments after school
	(1) Loans + grants - COA	(2) Loans + grants + earnings - COA	
Constrained x treated cohort	697	2163	1070
	(375.1)+	(925.4)*	(205.4)**
Dep var mean (cons, pre)	-$6,436	$15,006	$10,040
Observations	117,854	117,854	117,857

Notes: Point estimates from regressions of the outcome indicated in the column heading on an interaction between an indicator for being constrained (borrowing at the federal Stafford Loan limit) and an indicator for belonging to a treated cohort (academic year 2005 and later). Regressions also include entry cohort and entry program fixed effects and age, indicators for race/ethnicity, gender, college educated parents and enrollment in fall and spring, and constrained. Annual loan payments based on interest accumulation during enrollment and the assumptions that borrowers consolidate their loans when they enter repayment and pay on a 10-year amortization schedule (see Appendix B for additional details). In column (1), available resources equals the sum of loans and grants received during enrollment net of the cumulative cost of attendance. Column (2) definition is the same as column (1) with the addition of cumulative earnings during enrollment. Robust standard errors, clustered at the program level, in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

Table A.21: Heterogeneous effects of Grad PLUS on constrained students' outcomes by student demographic characteristics

	Cumulative loans (6 years)				Attainment + LM outcomes (10 years)		
	(1) PLUS	(2) Federal	(3) Private	(4) Total	(5) Any degree	(6) Any earnings	(7) ln(earnings
A. URM (N = 39,235)							
Constrained x treated cohort	4089	4239	-973	3311	0.030	0.023	0.016
	(948)**	(1683)*	(443)*	(1577)*	(0.014)*	(0.010)*	(0.033)
Dep var mean (cons, pre)	$251	$65,696	$2,221	$68,224	0.759	0.662	10.96
B. Not URM (N = 78,622)							
Constrained x treated cohort	7439	8235	-3678	4674	-0.001	0.010	0.050
	(1424)**	(1828)**	(1067)**	(1283)**	(0.008)	(0.010)	(0.027)+
Dep var mean (cons, pre)	$251	$68,668	$7,434	$76,398	0.855	0.583	11.30
C. Female (N = 71,387)							
Constrained x treated cohort	5237	5219	-2376	2919	0.002	0.020	0.041
	(1143)**	(1534)**	(784)**	(1151)*	(0.008)	(0.009)*	(0.027)
Dep var mean (cons, pre)	$188	$66,040	$4,756	$71,080	0.838	0.615	11.01
D. Male (N = 46,470)							
Constrained x treated cohort	7847	9148	-3494	5765	0.020	0.005	0.036
	(1436)**	(1844)**	(1055)**	(1456)**	(0.011)+	(0.012)	(0.031)
Dep var mean (cons, pre)	$323	$69,895	$7,414	$77,623	0.836	0.594	11.41
E. First generation (N = 45,150)							
Constrained x treated cohort	4143	3774	-1701	2215	0.008	0.012	0.019
	(844)**	(1450)**	(672)*	(1279)+	(0.010)	(0.012)	(0.030)
Dep var mean (cons, pre)	$311	$68,807	$3,444	$72,487	0.790	0.627	11.05
F. College educated parent(s) (N = 72,707)							
Constrained x treated cohort	7692	8782	-3654	5201	0.015	0.013	0.043
	(1465)**	(1805)**	(1020)**	(1276)**	(0.008)+	(0.010)	(0.028)
Dep var mean (cons, pre)	$219	$67,337	$7,399	$75,068	0.863	0.593	11.28
G. Below median EFC (N = 58,928)							
Constrained x treated cohort	5725	6762	-2870	3976	0.014	0.015	0.029
	(1493)**	(1988)**	(955)**	(1498)**	(0.009)	(0.009)	(0.029)
Dep var mean (cons, pre)	$229	$69,658	$5,120	$75,148	0.806	0.604	11.04
H. Above median EFC (N = 58,929)							
Constrained x treated cohort	6987	6748	-2947	3903	0.012	0.013	0.040
	(1249)**	(1704)**	(945)**	(1248)**	(0.009)	(0.011)	(0.027)
Dep var mean (cons, pre)	$269	$66,449	$6,697	$73,389	0.861	0.606	11.31
I. Out-of-state student (N = 16,989)							
Constrained x treated cohort	11170	12626	-5087	7567	0.041	0.038	0.092
	(2272)**	(2661)**	(1759)**	(2221)**	(0.021)*	(0.015)*	(0.072)
Dep var mean (cons, pre)	$420	$75,246	$10,799	$86,064	0.846	0.295	11.20
J. In-state student (N = 100,868)							
Constrained x treated cohort	5194	5430	-2430	3133	0.007	0.008	0.025
	(1113)**	(1510)**	(816)**	(1152)**	(0.007)	(0.009)	(0.023)
Dep var mean (cons, pre)	$213	$66,172	$4,915	$71,449	0.835	0.676	11.20

Notes: The sample includes first-time graduate students in the 2002 through 2008 entry cohorts who borrowed in their first year of enrollment. Each panel represents the results when restricting the sample further the specified subgroup of students. Point estimates from regressions of the outcome indicated in the column heading on an interaction between an indicator for being constrained (borrowing at the federal Stafford Loan limit) and an indicator for belonging to a treated cohort (academic year 2005 and later). Regressions also include entry cohort and entry program fixed effects and age, indicators for race/ethnicity, gender, college educated parents and enrollment in fall and spring, and constrained. Robust standard errors, clustered at the program level, in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

Table A.22: Heterogeneous effects of Grad PLUS on constrained students' outcomes by field of study and typical time to degree

		Cumulative loans (6 years)			Attainment & earnings (10 years)		
	(1) PLUS	(2) All federal	(3) Private	(4) All sources	(5) Any degree	(6) Any earnings	(7) ln(earnings)
A. STEM or medical degree program (N = 18,852)							
Constrained x treated cohort	2419	2982	-3437	-151	0.005	0.008	0.031
	(549)**	(3265)	(2325)	(2699)	(0.019)	(0.019)	(0.057)
Dep var mean (cons, pre)	$154	$83,986	$7,352	$91,431	0.844	0.493	11.17
B. Education program (N = 19,681)							
Constrained x treated cohort	202	-143	179	56	-0.006	0.010	0.096
	(73)**	(1558)	(119)	(1494)	(0.016)	(0.013)	(0.041)*
Dep var mean (cons, pre)	$136	$55,428	$138	$55,633	0.759	0.744	10.76
C. JD program (N = 11,638)							
Constrained x treated cohort	14927	20238	-2246	17292	0.011	-0.011	0.051
	(2624)**	(3990)**	(1950)	(4925)**	(0.020)	(0.017)	(0.077)
Dep var mean (cons, pre)	$232	$77,670	$12,645	$91,022	0.898	0.575	11.45
D. Business program (CIP code = 52) (N = 11,322)							
Constrained x treated cohort	1834	3096	-205	2905	0.021	0.010	-0.082
	(534)**	(1677)+	(224)	(1828)	(0.020)	(0.020)	(0.039)*
Dep var mean (cons, pre)	$185	$43,915	$907	$44,941	0.813	0.662	11.44
E. Other non-academic program (N = 24,258)							
Constrained x treated cohort	1834	3096	-205	2905	0.018	-0.014	0.081
	(534)**	(1677)+	(224)	(1828)	(0.016)	(0.016)	(0.050)
Dep var mean (cons, pre)	$185	$43,915	$907	$44,941	0.812	0.550	10.70
F. Median pre-period TTD <= 2 years (N = 94,690)							
Constrained x treated cohort	6793	6384	-2445	4045	0.008	0.013	0.040
	(1356)**	(1630)**	(799)**	(1188)**	(0.008)	(0.009)	(0.024)+
Dep var mean (cons, pre)	$200	$62,675	$6,102	$69,098	0.837	0.614	11.20
G. Median pre-period TTD > 2 years (N = 16,989)							
Constrained x treated cohort	2497	9906	-4237	5757	0.015	-0.010	-0.043
	(430)**	(3880)*	(2477)+	(3130)+	(0.023)	(0.024)	(0.049)
Dep var mean (cons, pre)	$859	$129,623	$4,881	$134,536	0.844	0.493	11.17

Notes: The sample includes first-time graduate students in the 2002 through 2008 entry cohorts who borrowed in their first year of enrollment. Each panel represents the results when restricting the sample to the subgroup of students who entered the type of program specified in the panel heading. Point estimates from regressions of the outcome indicated in the column heading on an interaction between an indicator for being constrained (borrowing at the federal Stafford Loan limit) and an indicator for belonging to a treated cohort (academic year 2005 and later). Regressions also include entry cohort and entry program fixed effects and age, indicators for race/ethnicity, gender, college educated parents and enrollment in fall and spring, and constrained. Robust standard errors, clustered at the program level, in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

Table A.23: Heterogeneous effects of Grad PLUS on constrained students' outcomes by type of institution

		Cumulative loans (6 years)			Attainment & earnings (10 years)		
	(1) PLUS	(2) All federal	(3) Private	(4) All sources	(5) Any degree	(6) Any earnings	(7) ln(earnings)
A. Health-related (N = 13,449)							
Constrained x treated cohort	3040	1236	-1	1437	0.012	0.009	0.048
	(550)**	(2875)	(688)	(3015)	(0.019)	(0.019)	(0.057)
Dep var mean (cons, pre)	$192	$96,435	$9,002	$105,928	0.844	0.493	11.17
B. Flagship (N = 15,630)							
Constrained x treated cohort	11346	11334	-4051	6898	-0.017	-0.017	0.037
	(3285)**	(3587)**	(847)**	(2765)*	(0.014)	(0.015)	(0.029)
Dep var mean (cons, pre)	$173	$62,916	$9,433	$72,800	0.946	0.520	11.63
C. R1 (N = 30,973)							
Constrained x treated cohort	5458	6585	-4082	2634	0.004	0.018	0.039
	(1829)**	(2765)*	(1909)*	(2041)	(0.010)	(0.012)	(0.043)
Dep var mean (cons, pre)	$179	$61,406	$5,023	$66,563	0.837	0.623	11.18
D. Other (N = 57,805)							
Constrained x treated cohort	5867	6464	-2039	4675	0.024	0.023	0.037
	(1698)**	(2217)**	(1243)	(1808)*	(0.012)+	(0.013)+	(0.029)
Dep var mean (cons, pre)	$345	$70,276	$4,598	$75,182	0.774	0.631	11.01

Notes: The sample includes first-time graduate students in the 2002 through 2008 entry cohorts who borrowed in their first year of enrollment. Each panel represents the results when restricting the sample to students who entered the type of institution specified in the panel heading. Point estimates from regressions of the outcome indicated in the column heading on an interaction between an indicator for being constrained (borrowing at the federal Stafford Loan limit) and an indicator for belonging to a treated cohort (academic year 2005 and later). Regressions also include entry cohort and entry program fixed effects and age, indicators for race/ethnicity, gender, college educated parents and enrollment in fall and spring, and constrained. Robust standard errors, clustered at the program level, in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

Table A.24: Heterogeneity in the effect of Grad PLUS on constrained students' outcomes by Great Recession severity

	Cumulative loans (6 years)				Attainment + LM outcomes (10 years)		
	(1) PLUS	(2) Federal	(3) Private	(4) All sources	(5) Any degree	(6) Any earnings	(7) ln(earnings)
Constrained x treated cohort	6536	7547	-2609	4878	0.020	0.011	0.042
	(1703)**	(2117)**	(1195)*	(1566)**	(0.013)	(0.011)	(0.034)
x Δ UR 2007-2009	3164	5153	-1389	3809	-0.0004	0.027	-0.042
	(1034)**	(2090)*	(929)	(2046)+	(0.018)	(0.018)	(0.050)
Observations	81,395	81,395	81,395	81,395	81,395	81,395	53,695

Notes: The sample includes first-time graduate students in the 2002 through 2008 entry cohorts who borrowed in their first year of enrollment. Δ UR is the percentage point change in the county unemployment rate between 2007 and 2009 for the county in which the graduate program a student first entered was located (as a measure of the severity of the Great Recession). Point estimates from regressions of the outcome indicated in the column heading on an interaction between an indicator for being constrained (borrowing at the federal Stafford Loan limit) and an indicator for belonging to a treated cohort (academic year 2005 and later) and the triple interaction between constrained status, treated cohort, and Δ UR. Regressions also include entry cohort and entry program fixed effects and age, indicators for race/ethnicity, gender, college educated parents and enrollment in fall and spring, and constrained and interactions between these variables and Δ UR. Robust standard errors, clustered at the program level, in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

Table A.25: Robustness of effects of Grad PLUS on constrained students' outcomes to excluding partially treated entry cohorts

	Cumulative loans (6 years)				Attainment + LM outcomes (10 years)		
	(1) PLUS	(2) Federal	(3) Private	(4) All sources	(5) Any degree	(6) Any earnings	(7) ln(earnings)
Constrained x treated cohort	6432	7267	-2538	4679	0.020	0.009	0.045
	(1713)**	(2194)**	(1173)*	(1649)**	(0.012)	(0.011)	(0.034)
Observations	81,395	81,395	81,395	81,395	81,395	81,395	53,695
Dep var mean (cons, pre)	$139	$63,978	$4,323	$68,512	0.821	0.597	11.15

Notes: The sample includes first-time graduate students in the 2002-2003 and 2006-2008 entry cohorts who borrowed in their first year of enrollment. Students in partially treated entry cohorts (2004 and 2005) are excluded. Point estimates from regressions of the probability of the type of degree received (indicated in the panel heading) by the specified number of years since entry on an interaction between an indicator for being constrained (borrowing at the federal Stafford Loan limit) and an indicator for belonging to a treated cohort (academic year 2005 and later). Regressions also include entry cohort and entry program fixed effects and age, indicators for race/ethnicity, gender, college educated parents and enrollment in fall and spring, and constrained. Robust standard errors, clustered at the program level, in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

Appendix B: Data Appendix

This appendix provides additional information on the construction of the analysis data set and selected variables.

B.1 Texas Higher Education Coordinating Board and Texas Workforce Commission data

The Texas Higher Education Coordinating Board (THECB) financial aid data has increasing coverage of students over time. Information on enrollment and attainment within public institutions is available starting in 2001 and in nonprofit institutions starting in 2003. All students who enrolled in these institutions are observed.

The THECB financial aid data (which also contains information on cost of attendance) does not contain records for all students. From 2003 to 2006, it includes students who received aid that required a needs calculation. From 2007 to 2009, it includes any student who filed a federal student aid application (FAFSA) even if they did not receive aid. From 2010 to the present, it also includes students who did not file a FAFSA and received merit aid. Some of the fields in the financial aid data report aid received through multiple programs (that may or may not overlap in the years they were in operation) in the same field. Notably, the field used to report Grad PLUS loan aid was also used for reporting Supplemental Loans for Students (SLS) aid. However, the SLS program is very small.

The Texas Workforce Commission (TWC) data contain quarterly earnings records for all workers covered by the Texas Unemployment Insurance (UI) system. UI records cover employers who pay at least $1500 in gross wages to employees or have at least one employee during twenty different weeks in a calendar year. Students employed by their college or university are not included in these data, but work study funding is observable in the THECB data. When constructing measures of earnings or the probability of having any earnings, we align quarters with academic years. I.e., AY 2010 earnings is the sum of earnings from 2009-Q4 through 2010-Q3.

To construct an analysis sample of first-time entering students, we exclude students who appear to enter graduate school in the first year of available data (2001 for publics, 2003 for nonprofits). This is to ensure we correctly identify new students since we need at least one year of data to identify if a student had previously enrolled in a graduate program. Because our program-level measure of cost of attendance comes from student records in the financial aid data, we create an alternative financial aid sample by limiting the underlying financial aid records to students who meet the strictest sample definition (e.g., the 2003 through 2006 inclusion criteria – students who received aid that required a need based calculation).

B.2 Defining programs of study

As discussed in Section 3, in some cases, students entering a doctoral program are classified as master's degree seeking, even if they are pursuing a PhD. Because not all doctoral degree-seeking

students complete their program, we cannot use the degree received to assign students to programs. Instead, we focus on the set of students who do receive a degree from the program and calculate the percentage of such students who receive a terminal master's degree. We classify a student as receiving a terminal master's degree if they receive a master's degree but no additional degrees in that CIP by institution combination. For each 4-digit CIP code, we then calculate the percentage of degrees received that were terminal master's degrees. We then define a program as a professional (i.e., nonacademic) degree program if more than 85 percent of degrees granted were terminal master's degrees. Otherwise, a program is classified as academic.

To provide an example, if over 85 percent of degrees awarded to graduates with a CIP code of 4506 (Economics) at the University of Texas at Austin were master's degrees (not followed by any subsequent doctoral degree in the same CIP code), this would be classified as a professional master's degree in economics, but if less than 85 percent of degrees awarded were terminal masters, all students would be classified as academic doctoral students (even if their highest degree obtained was a master's degree). Students in a program with a 2-digit CIP code of 51 (Health Professions) or 13 (education) are considered to be in a professional degree program even if most of the degrees granted are doctoral in the data. All health and education programs are classified as non-academic.

In most cases, programs are defined by 4-digit CIP but because the level of disaggregation is not constant across CIP codes (i.e., economics is defined at the 4-digit CIP-level but psychology is defined at the 2-digit CIP-level), we deviate from this approach in a handful of cases. We combine 2-digit CIP codes 14 and 15 into a single category of engineering. All programs with a 2-digit CIP code of 13 are combined into a single category of education. All institution by 4-digit CIP code combinations that do not meet the 85 percent terminal master's degree cut-off are combined into a single category of academic programs. We exclude students listed as being in a medical residency as their field of study.

A small number of CIP codes are added, deleted, or combined every decade. We use crosswalks provided by the National Center for Education Statistics (NCES) to adjust observed CIP codes to ensure that fields of study are measured consistently over the years our data span.[1] Specifically, we map all earlier CIP codes to the 2010 code. If a CIP code did not appear in 2010 but did appear in 1990 and 2000, we used the codes from 1990 and 2000. In one instance, (14.17) the 1990 code was more aggregated than subsequent codes and so we kept those codes aggregated. The 1990 4-digit CIP codes of 26.06 and 51.13 mapped into a number of 2000 CIPs. To handle these, we recoded all underlying 6-digit CIPs into their 2000 6-digit CIP, before re-applying general aggregation methods as above.

[1] See https://nces.ed.gov/pubs2002/cip2000/crosswalk.ASP and https://nces.ed.gov/ipeds/cipcode/crosswalk.aspx.

The Texas Higher Education Coordinating Board requires different information from private nonprofit universities (which they call "Independent Colleges and Universities") than from public universities. In many cases, we do not observe CIP codes for enrolled students and can only identify programs with separate tuition classifications: theology, education, law, and chiropractic programs. Graduates from these programs made up 22 percent of all graduate degrees granted by nonprofit institutions in 2006 and overall, we can identify programs for 82 percent of all graduate student degree recipients in 2006 (authors' calculations using IPEDS completions data).

Our main program-level estimates are robust to using alternative terminal master's degree cut-offs in our classification of academic and professional programs. Table B.4 compares estimated effects on program access using our main definition (Panel A, replicating estimates in Panel A of Table 3) and the samples that result when academic programs are defined using am 80 percent terminal master's degree cut-off (Panel B) and a 75 percent threshold (Panel C). Likewise, Table B.5 shows that estimated effects of exposure to Grad PLUS (i.e., percent of students constrained at baseline) are robust to these alternate definitions.

B.3 Predicting student loan payments

The THECB data does not include information on loan repayment or interest rates for state and private student loans. To predict a student's loan repayments, we must impose a number of assumptions. We first calculate interest accrual while a student is still enrolled. For federal loans, we use the type of loan and academic year of disbursement to assign interest rates based on Hegji (2021). We impute interest rates for private student loans based on the differential between the median private student loan interest rate, for loans issued to graduate students, reported by the Consumer Financial Protection Bureau (2012) and the Stafford Loan interest rate. Stafford Loans issued before 2007 had a variable interest rate that depended on whether a student was enrolled or in repayment. Thus, for private loans issued before 2007, we calculate the differential between the median private student loan interest rate in the CFPB report and the average in-school Stafford Loan rate (4 percentage points) and assign private loans issued in a given academic year the Stafford in-school interest rate for that year plus 4 percentage points. Stafford Loans issued after 2007 had a fixed interest rate that depended only on the academic year it was disbursed. We calculate the differential between the median private student loan interest rate and the average Stafford Loan fixed interest rate after 2007 (1 percentage point) and assign private loans issued after 2007 the prevailing Stafford Loan interest rate plus 1 percentage point. Federal Perkins loans do not accrue interest during enrollment and have a 5 percent interest rate during repayment. We assume that loans issued by the state also do not accrue interest during enrollment and have a 4.2 percent interest rate based on the largest loan programs offered by Texas.

Given these interest rates, we calculate interest accrued during enrollment and assume that students consolidate all of their loans when they enter repayment. We assume that once loans are consolidated, the (balance-weighted) average interest rate is fixed over the duration of repayment. We assume that students repay their loans on a 10-year amortization schedule and pay the same amount each month.

Based on these assumptions, constrained borrowers in entry cohorts that did not gain access to Grad PLUS accumulated approximately $4,200 in interest and entered repayment with approximately $78,200 in outstanding debt, on average. Constrained borrowers in cohorts that did gain access to Grad PLUS, accumulated approximately $5400 in interest and entered repayment with approximately $83,500 in outstanding debt. At repayment entry, constrained borrowers in untreated cohorts faced a 4.7 percent interest rate, on average, and constrained borrowers in treated entry cohorts faced a 5 percent interest rate.

B.4 Additional tables

Table B.1: Texas data coverage for observed programs

	(1) Pre-Grad PLUS (2004)	(2) Post Grad PLUS (2008)
Public	0.51	0.47
Nonprofit	0.41	0.40
Nonprofit, covered programs	0.21	0.19
For-profit	0.05	0.12
Observed in TX data*	0.72	0.67

Notes: Programs that would be observed in the Texas administrative data are all graduate programs in public and institutions and theology, education, law, and chiropractic programs in private nonprofit institutions. These estimates are based on the distribution of entering graduate students across institutions and programs of study in the 2004 and 2008 waves of the National Postsecondary Student Aid Study (via PowerStats). The codes to retrieve 2004 results are: bngyht, bkedjc. The codes to retrieve 2008 results are: zdbsbc, tnpmqj.

Table B.2: Characteristics of programs that would be observed and unobserved in Texas data

	Pre-Grad PLUS (2004)		Post-Grad PLUS (2008)	
	(1) Observed	(2) Not observed	(3) Observed	(4) Not observed
Weighted sample size	699,138	235,785	890,739	348,509
Female	0.59	0.54	0.62	0.58
Race/ethnicity/nativity*				
Asian	0.10	0.10	0.10	0.08
Black	0.09	0.14	0.12	0.28
Hispanic	0.10	0.09	0.08	0.05
White	0.67	0.65	0.68	0.55
International student	0.07	‡	0.08	‡
Age (in yrs.)	30	32	31	33
Borrowers	0.40	0.55	0.42	0.58
Constrained borrowers	0.12	0.25	0.11	0.22
Amount borrowed (nominal)				
Total loans	$5,977	$8,819	$7,217	$9,973
Federal Stafford loans	$5,086	$7,570	$5,526	$7,992
Federal PLUS loans	--	--	$795	$488
Federal Perkins loans	$143	$153	$104	‡
State loans	‡	$0	‡	‡
Private and other loans	$634	$1,053	$791	$1,434
Total grants (nominal)	$2,085	$2,912	$2,887	$3,510
Cost of attendance (nominal)	$17,052	$22,600	$20,409	$26,469

Notes: See Table A.1 for definition of observed and unobserved programs. * Race/ethnicity is for all students, including international students (in Texas data, race/ethnicity is not observed for international students). ‡ Didn't meet NPSAS reporting standards. Constrained borrowers are those with annual federal student loans at the statutory limit. These estimates are based on the nationally representative sample of entering graduate students in the 2004 and 2008 waves of the National Postsecondary Student Aid Study (via PowerStats). The codes to retrieve 2004 results are: tykqbj, grklqp, bsagot, nqdzsk. The codes to retrieve 2008 results are: sznmjt, bagrvd, ngxiia.

Table B.3

Flagship	Health-related	R1	Other
The University of Texas at Austin	The University of Texas Southwestern Medical Center	University of North Texas	Sul Ross State University Rio Grande College
Texas A&M University	The University of Texas Health Science Center at San Antonio	Texas Tech University	Abilene Christian University
	Baylor College of Medicine	University of Houston	Angelo State University
	The Texas A&M University System Health Science Center	The University of Texas at Arlington	Austin College
	University of North Texas Health Science Center at Fort Worth	The University of Texas at El Paso	Concordia University Texas
	Texas Tech University Health Sciences Center	The University of Texas at Dallas	East Texas Baptist University
	The University of Texas Health Science Center at Houston	The University of Texas at San Antonio	Texas A&M University-Commerce
	The University of Texas Medical Branch at Galveston		Hardin-Simmons University
			Howard Payne University
			Houston Christian University
			University of the Incarnate Word
			Lamar University
			Lubbock Christian University
			University of Mary Hardin-Baylor
			Midwestern State University
			The University of Texas-Rio Grande Valley
			Sam Houston State University
			Schreiner University
			Southern Methodist University
			Texas State University
			Southwestern Adventist University
			St. Edward's University
			St. Mary's University
			Stephen F. Austin State University
			Sul Ross State University
			Prairie View A&M University
			Tarleton State University
			Texas Chiropractic College
			Texas Christian University
			Texas A&M University-Kingsville
			Texas Southern University
			Texas Wesleyan University
			Texas Women's University
			University of St. Thomas
			Wayland Baptist University
			West Texas A&M University
			South Texas College of Law
			Texas A&M International University
			The University of Texas of the Permian Basin
			Texas A&M University at Galveston
			Texas A&M University-Corpus Christi
			The University of Texas at Tyler
			University of Houston-Clear Lake
			University of Houston-Downtown
			University of Houston-Victoria
			Parker University
			Texas A&M University-Texarkana
			The University of Texas at Brownsville
			Texas A&M University-Central Texas
			University of North Texas at Dallas
			Texas A&M University-San Antonio

Table B.4: Robustness of estimated effects on program enrollment and composition

	(1)	Percent (0-100%) of entering students who are:						
	Enrollment	(2) Black	(3) Hispanic	(4) API	(5) White	(6) Internl	(7) Men	(8) First gen
A. 85% terminal MA cut-off (main results)								
Projected limit increase ($1k) x pc	-1.0	-0.01	0.02	0.02	0.08	-0.03	-0.004	0.12
	(0.8)	(0.02)	(0.03)	(0.02)	(0.09)	(0.06)	(0.03)	(0.09)
Observations	2,380	2,380	2,380	2,380	2,380	2,380	2,380	2,380
B. 80% terminal MA cut-off for nonacademic								
Projected limit increase ($1k) x pc	-1.1	-0.02	0.02	0.02	0.08	-0.04	0.00	0.11
	(0.8)	(0.02)	(0.03)	(0.02)	(0.09)	(0.07)	(0.03)	(0.09)
Observations	2,576	2,576	2,576	2,576	2,576	2,576	2,576	2,571
C. 75% terminal MA cut-off for nonacademic								
Projected limit increase ($1k) x pc	-1.0	-0.02	0.01	0.02	0.08	-0.03	-0.01	0.11
	(0.8)	(0.02)	(0.03)	(0.02)	(0.09)	(0.07)	(0.04)	(0.09)
Observations	2,779	2,779	2,779	2,779	2,779	2,779	2,779	2,773

Notes: The sample includes graduate programs that had entering students in each of the 2004 through 2010 academic years and had at least 20 entering students who filed a FAFSA in 2004 through 2006. Each panel contains point estimates from separate regressions of first-year enrollment or percentage of entering students (0-100) with the given demographic characteristic on an interaction between post-Grad PLUS entry cohort and the projected federal loan limit increase based on the program's 2006 average cost of attendance (COA). All specifications also include entry cohort and program fixed effects. Columns (2) through (7) estimates weighted by the size of the entering cohort. Robust standard errors, clustered at the program level, in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

Table B.5: Robustness of estimated effects on borrowing and prices

	(3) COA	(4) FTFY COA	(1) Grad PLUS loans	(2) Total Fed. loans	(5) Grants	(6) Net price
A. 85% terminal MA cut-off for academic programs (main estimates)						
% constrained*Post	66.4	62.7	88.1	60.4	25.6	38.8
	(23.5)**	(18.9)**	(18.4)**	(32.0)+	(15.2)+	(33.5)
Observations	2,336	2,336	2,336	2,336	2,336	2,336
B. 80% terminal MA cut-off for academic program def						
% constrained*Post	65.7	62.1	88.1	60.2	25.7	38.1
	(23.4)**	(19.0)**	(18.4)**	(32.1)+	(15.2)+	(33.5)
Observations	2,576	2,576	2,576	2,576	2,576	2,576
C. 75% terminal MA cut-off for academic program def						
% constrained*Post	64.9	61.7	88.2	60.2	25.7	37.3
	(23.4)**	(19.0)**	(18.5)**	(32.1)+	(15.3)+	(33.6)
Observations	2,728	2,728	2,728	2,728	2,728	2,728

Notes: The sample includes a balanced panel graduate programs with enrollment in the 2003 through 2010 academic years and at least 20 federal aid recipients enrolled per year, on average, between 2003 and 2006. Each panel contains point estimates from separate regressions of the outcome indicated in the column heading on an interaction between post-Grad PLUS academic year and exposure to Grad PLUS (the percent of students with Stafford Loans at or above the Stafford maximum (adjusted for part-year enrollment), on average, between 2004 and 2006). Observations are weighted by baseline program size except where indicated. Robust standard errors, clustered at the program level, in parentheses; ** $p < 0.01$, + $p < 0.1$.